Extracting Recipes from Cooking Videos using V-JEPA2 Embeddings

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Introduction

V-JEPA 2 provides an accessible way to encode complex physical interactions in videos without fine-tuning the visual backbone.
Understanding **egocentric cooking videos** is challenging: first-person footage captures natural kitchen activities but suffers from clutter, and difficult recognition of objects.

We propose a **two-stage pipeline**:

1. **Relevance filtering** — separate cooking actions from background events.
2. **Action classification** — fine-grained verb/noun recognition on relevant clips.

We exploit **V-JEPA2 embeddings** and train a lightweight temporal classifier on top, keeping compute requirements minimal.

Dataset: EPIC-KITCHENS-100

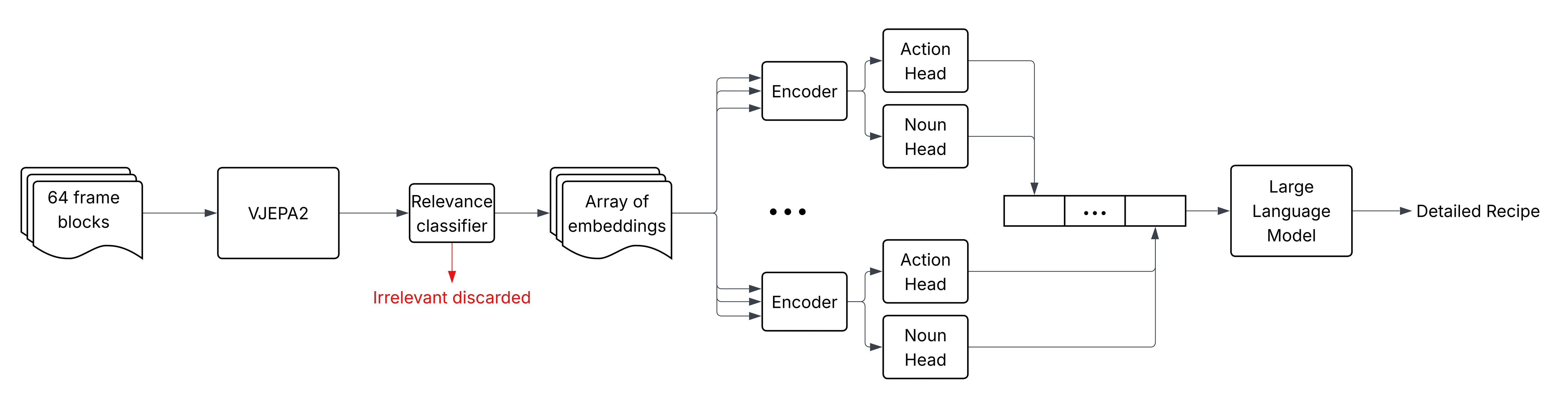
- **407 embedded videos**, split into **170,007** temporal blocks
- **1024-D V-JEPA2 embeddings** per 64-frame block
- **116,812** blocks overlap an annotation; **53,195** are background/unannotated
- **5,988** relevant verb–noun action-pair classes used by the action head

Block label	Fraction
Relevant	60.0%
Irrelevant/background	40.0%

Why V-JEPA2-small?

We use the **small** variant of V-JEPA2 (ViT-L backbone, ≈307M parameters). Its compact footprint makes local inference feasible on a **MacBook Pro** or consumer laptop with sufficient GPU RAM. For this project we used **cloud compute** to embed the dataset in reasonable time, then trained the classifier on Apple MPS.

Pipeline Overview



Classification Heads

A shared temporal encoder reads a 7-block window (radius 3). Each 1024-D embedding is projected to 768 dimensions, learned positional encodings are added, and a 2-layer, 8-head TransformerEncoder aggregates context. Missing neighbors at video boundaries are zero-padded and masked.

The center token feeds four MLP heads: **relevance** (binary), **verb**, **noun**, and **action pair**. Relevance is trained on all blocks; action losses are masked to valid relevant blocks.

At inference, action prediction combines action, verb, and noun log probabilities with a smoothed verb–noun pair prior ($w = 0.25$). A 3-seed ensemble (42, 123, 999) averages logits before decoding.

Why a Transformer Encoder?

A single block provides only a narrow temporal context. A Transformer encoder attends across the full sequence of block embeddings, aggregating multi-window evidence before the center-token heads decide relevance and action labels.

Results

Metric	Val	Test
Relevance accuracy	82.9%	84.0%
Relevance F1	86.1%	87.0%
Verb accuracy	74.4%	77.3%
Noun accuracy	77.7%	80.3%
Exact action	73.0%	75.8%

Action scores are measured after the relevance gate on true relevant blocks. With oracle relevance, test exact-action accuracy reaches **84.4%**; on predicted relevant blocks it is **87.2%**.

LLM Post-Processing

The action classifier outputs a **sequence of (verb, noun) pairs** aligned to video blocks. We pass this sequence to **Llama 3.2** which synthesizes the raw event list into a fluent, human-readable recipe: coherent steps, resolved repetitions, and natural prose.

This bridges the gap between frame-level predictions and interpretable cooking instructions.

Conclusion

- **Two-stage design** (relevance + classification) cuts background noise before fine-grained prediction.
- **Transformer encoder** provides multi-block temporal context with minimal added parameters.
- **LLM post-processing** turns sparse event sequences into readable recipes.
- Pipeline runs on consumer hardware with no backbone tuning.